

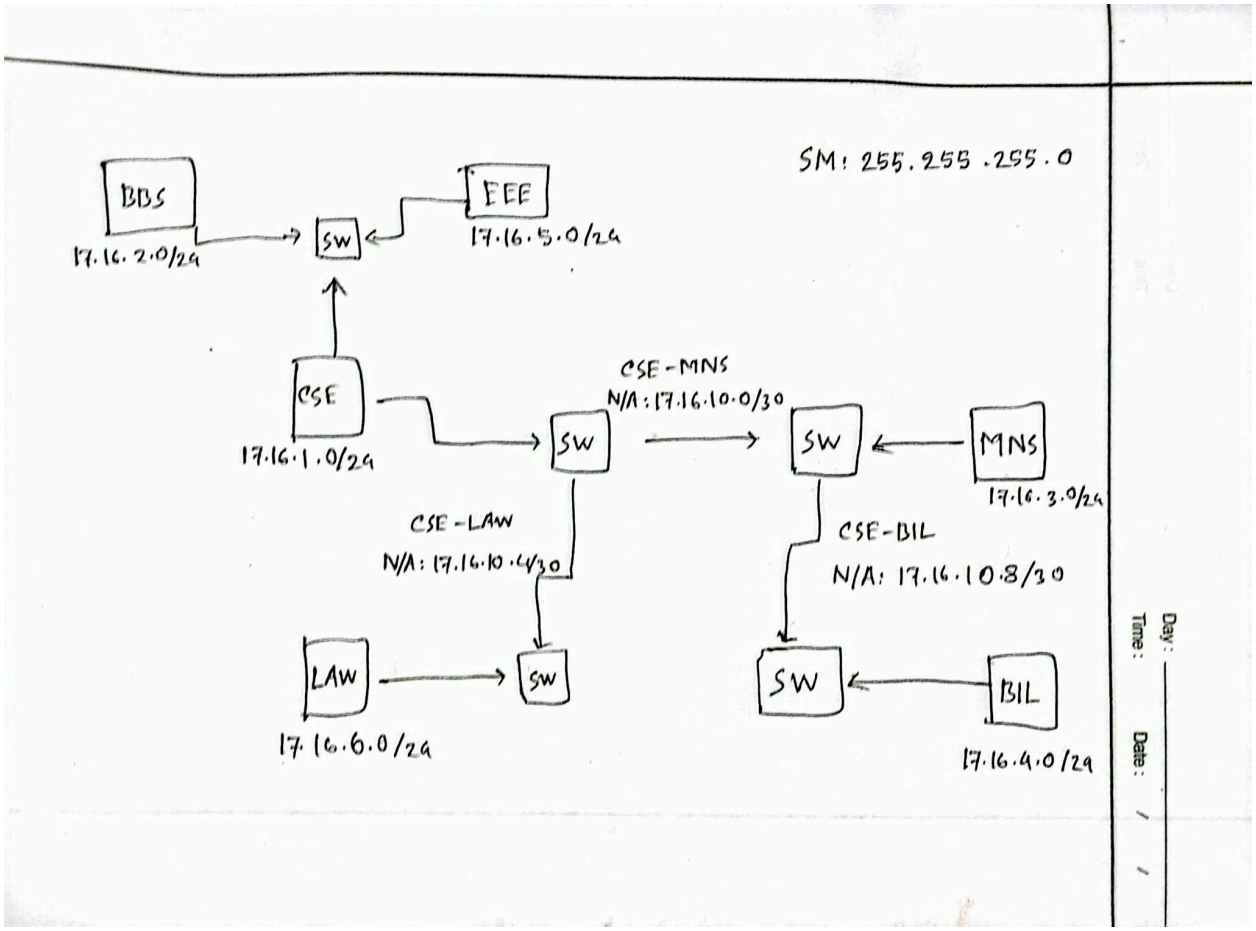
CSE421 Project

Topic 07: BRACU Network

Group: 2206

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VLSM Tree


SM: 255.255.255.0
BBS
17.16.2.0/24
SW
EEE
17.16.5.0/24
CSE
17.16.1.0/24
SW
CSE-MNS
N/A: 17.16.10.0/30
SW
MNS
17.16.3.0/24
CSE-LAW
N/A: 17.16.10.4/30
CSE-BIL
N/A: 17.16.10.8/30
LAW
17.16.6.0/24
SW
SW
BIL
17.16.4.0/24
Day:
Time:
Date:

IP address table

Device Name	Interface / Type	IP Address	Subnet Mask	Default Gateway	Description
R-CSE	GigabitEthernet0/0	17.16.1.1	255.255.255.0	N/A	CSE LAN Gateway
R-CSE	GigabitEthernet0/1	17.16.20.1	255.255.255.0	N/A	To SW-Core
R-CSE	GigabitEthernet0/2	17.16.10.1	255.255.255.252	N/A	To R-MNS
R-CSE	Serial0/0/0	17.16.10.5	255.255.255.252	N/A	To R-LAW
R-CSE	Serial0/0/1	17.16.10.9	255.255.255.252	N/A	To R-BIL
R-MNS	GigabitEthernet0/1	17.16.3.1	255.255.255.0	N/A	MNS LAN Gateway
R-MNS	GigabitEthernet0/0	17.16.10.2	255.255.255.252	N/A	To R-CSE
R-EEE	GigabitEthernet0/0	17.16.5.1	255.255.255.0	N/A	EEE LAN Gateway
R-EEE	GigabitEthernet0/1	17.16.20.2	255.255.255.0	N/A	To SW-Core
R-BBS	GigabitEthernet0/0	17.16.2.1	255.255.255.0	N/A	BBS LAN Gateway

R-BBS	GigabitEthernet0/1	17.16.20.3	255.255.255.0	N/A	To SW-Core
R-LAW	GigabitEthernet0/0	17.16.6.1	255.255.255.0	N/A	LAW LAN Gateway
R-LAW	Serial0/0/0	17.16.10.6	255.255.255.252	N/A	To R-CSE
R-BIL	GigabitEthernet0/1	17.16.4.1	255.255.255.0	N/A	BIL LAN Gateway
R-BIL	Serial0/0/0	17.16.10.10	255.255.255.252	N/A	To R-CSE
SRV-CSE	FastEthernet0	17.16.1.10	255.255.255.0	17.16.1.1	Email, Web, DNS Server
SRV-MNS	FastEthernet0	17.16.3.10	255.255.255.0	17.16.3.1	Email, Web Server
SRV-EEE	FastEthernet0	17.16.5.10	255.255.255.0	17.16.5.1	Email Server
SRV-BBS	FastEthernet0	17.16.2.10	255.255.255.0	17.16.2.1	Email Server
SRV-LAW	FastEthernet0	17.16.6.10	255.255.255.0	17.16.6.1	Email, DHCP Server
SRV-BIL	FastEthernet0	17.16.4.10	255.255.255.0	17.16.4.1	Email, DHCP Server
PC-CSE1	FastEthernet0	DHCP (17.16.1.51)	255.255.255.0	17.16.1.1	DHCP Client from R-CSE
PC-CSE2	FastEthernet0	DHCP	255.255.255.0	17.16.1.1	DHCP Client from R-CSE
PC-BBS1	FastEthernet0	DHCP	255.255.255.0	17.16.2.1	DHCP Client from R-CSE

PC-BBS2	FastEthernet0	DHCP	255.255.255.0	17.16.2.1	DHCP Client from R-CSE
PC-MNS1	FastEthernet0	DHCP	255.255.255.0	17.16.3.1	DHCP Client from R-CSE
PC-MNS2	FastEthernet0	DHCP	255.255.255.0	17.16.3.1	DHCP Client from R-CSE
PC-BIL1	FastEthernet0	DHCP	255.255.255.0	17.16.4.1	DHCP Client from SRV-BIL
PC-BIL2	FastEthernet0	DHCP	255.255.255.0	17.16.4.1	DHCP Client from SRV-BIL
PC-EEE1	FastEthernet0	DHCP	255.255.255.0	17.16.5.1	DHCP Client from R-CSE
PC-EEE2	FastEthernet0	DHCP	255.255.255.0	17.16.5.1	DHCP Client from R-CSE
PC-LAW1	FastEthernet0	DHCP	255.255.255.0	17.16.6.1	DHCP Client from SRV-LAW
PC-LAW2	FastEthernet0	DHCP	255.255.255.0	17.16.6.1	DHCP Client from SRV-LAW
PRN-CSE	FastEthernet0	17.16.1.20	255.255.255.0	17.16.1.1	Network Printer
PRN-BBS	FastEthernet0	17.16.2.20	255.255.255.0	17.16.2.1	Network Printer
PRN-MNS	FastEthernet0	17.16.3.20	255.255.255.0	17.16.3.1	Network Printer

PRN-BIL	FastEthernet0	17.16.4.20	255.255.255.0	17.16.4.1	Network Printer
PRN-EEE	FastEthernet0	17.16.5.20	255.255.255.0	17.16.5.1	Network Printer
PRN-LAW	FastEthernet0	17.16.6.20	255.255.255.0	17.16.6.1	Network Printer

Router configuration command

- **R-CSE :**

```
! --------------------------------------------
! R-CSE Router - Central Router Configuration
! --------------------------------------------
hostname R-CSE

! Interface Configurations
interface GigabitEthernet0/0
 description CSE-LAN to SW-CSE
 ip address 17.16.1.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 description To SW-Core (CSE-EEE-BBS Core)
```

```
 ip address 17.16.20.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/2
 description To R-MNS (Point-to-Point)
 ip address 17.16.10.1 255.255.255.252
 no shutdown

interface Serial0/0/0
 description To R-LAW (Point-to-Point)
 ip address 17.16.10.5 255.255.255.252
 clock rate 64000
 no shutdown

interface Serial0/0/1
 description To R-BIL (Point-to-Point)
 ip address 17.16.10.9 255.255.255.252
 clock rate 64000
 no shutdown

! DHCP Configuration
ip dhcp excluded-address 17.16.1.1 17.16.1.50
ip dhcp excluded-address 17.16.2.1 17.16.2.50
ip dhcp excluded-address 17.16.3.1 17.16.3.50
ip dhcp excluded-address 17.16.5.1 17.16.5.50

ip dhcp pool CSE-PCs
 network 17.16.1.0 255.255.255.0
 default-router 17.16.1.1
 dns-server 17.16.1.10

ip dhcp pool BBS-PCs
 network 17.16.2.0 255.255.255.0
 default-router 17.16.2.1
 dns-server 17.16.1.10

ip dhcp pool MNS-PCs
 network 17.16.3.0 255.255.255.0
 default-router 17.16.3.1
 dns-server 17.16.1.10
```

```
ip dhcp pool EEE-PCs
 network 17.16.5.0 255.255.255.0
 default-router 17.16.5.1
 dns-server 17.16.1.10

! Static Routes for Direct Connections
ip route 17.16.3.0 255.255.255.0 17.16.10.2
ip route 17.16.6.0 255.255.255.0 17.16.10.6
ip route 17.16.4.0 255.255.255.0 17.16.10.10
ip route 17.16.5.0 255.255.255.0 17.16.20.2
ip route 17.16.2.0 255.255.255.0 17.16.20.3

! RIP Routing for Core Network
router rip
 version 2
 no auto-summary
 network 17.16.1.0
 network 17.16.20.0
 network 17.16.10.0

line con 0
line vty 0 4
end
```

- **R-MNS:**

```
! --------------------------------------------
! R-MNS Router Configuration
! --------------------------------------------
hostname R-MNS

! Interface Configurations
interface GigabitEthernet0/1
 description MNS-LAN to SW-MNS
 ip address 17.16.3.1 255.255.255.0
```

```
 no shutdown

interface GigabitEthernet0/0
 description To R-CSE (Point-to-Point)
 ip address 17.16.10.2 255.255.255.252
 no shutdown

! Default Route to CSE (as per requirements)
ip route 0.0.0.0 0.0.0.0 17.16.10.1

line con 0
line vty 0 4
end
```

- **R-EEE:**

```
! ---------------------------------------------
! R-EEE Router Configuration
! ---------------------------------------------
hostname R-EEE

! Interface Configurations
interface GigabitEthernet0/0
 description EEE-LAN to SW-EEE
 ip address 17.16.5.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 description To SW-Core
 ip address 17.16.20.2 255.255.255.0
 no shutdown

! Default Route to CSE
ip route 0.0.0.0 0.0.0.0 17.16.20.1

! RIP Routing
router rip
 version 2
```

```
 no auto-summary
 network 17.16.5.0
 network 17.16.20.0

line con 0
line vty 0 4
end
```

- **R-BBS:**

```
! --------------------------------------------
! R-BBS Router Configuration
! --------------------------------------------
hostname R-BBS

! Interface Configurations
interface GigabitEthernet0/0
 description BBS-LAN to SW-BBS
 ip address 17.16.2.1 255.255.255.0
 no shutdown

interface GigabitEthernet0/1
 description To SW-Core
 ip address 17.16.20.3 255.255.255.0
 no shutdown

! Default Route to CSE
ip route 0.0.0.0 0.0.0.0 17.16.20.1

! RIP Routing
router rip
 version 2
 no auto-summary
 network 17.16.2.0
 network 17.16.20.0

line con 0
line vty 0 4
```

```
end
```

- **R-LAW:**

```
! --------------------------------------------
! R-LAW Router Configuration
! --------------------------------------------
hostname R-LAW

! Interface Configurations
interface GigabitEthernet0/0
 description LAW-LAN to SW-LAW
 ip address 17.16.6.1 255.255.255.0
 no shutdown

interface Serial0/0/0
 description To R-CSE (Point-to-Point)
 ip address 17.16.10.6 255.255.255.252
 no shutdown

! Static Routes to reach all networks via CSE
ip route 17.16.0.0 255.255.0.0 17.16.10.5

! Recursive route via BIL (if CSE-LAW link down)
ip route 17.16.1.0 255.255.255.0 17.16.4.1 200
ip route 17.16.2.0 255.255.255.0 17.16.4.1 200
ip route 17.16.3.0 255.255.255.0 17.16.4.1 200
ip route 17.16.5.0 255.255.255.0 17.16.4.1 200

line con 0
line vty 0 4
End
```

- **R-BIL:**

```
! --------------------------------------------
! R-BIL Router Configuration
! --------------------------------------------
hostname R-BIL

! Interface Configurations
interface GigabitEthernet0/1
 description BIL-LAN to SW-BIL
 ip address 17.16.4.1 255.255.255.0
 no shutdown

interface Serial0/0/0
 description To R-CSE (Point-to-Point)
 ip address 17.16.10.10 255.255.255.252
 no shutdown

! Static Routes to reach all networks via CSE
ip route 17.16.0.0 255.255.0.0 17.16.10.9

! Recursive route via LAW (if CSE-BIL link down)
ip route 17.16.1.0 255.255.255.0 17.16.6.1 200
ip route 17.16.2.0 255.255.255.0 17.16.6.1 200
ip route 17.16.3.0 255.255.255.0 17.16.6.1 200
ip route 17.16.5.0 255.255.255.0 17.16.6.1 200

line con 0
line vty 0 4
end
```